Stanley Chueh 闕楷宸

ROBOTICS & EMBODIED AI — VLA · ROBOT LEARNING · SIM-TO-REAL

t114318047@ntut.org.tw · 0978 455 076
Taipei, Taiwan · stanleychueh.com
github.com/StanleyChueh · linkedin.com/in/stanley-chueh

Robotics researcher building learning-based manipulation and navigation systems — from teleoperated data collection and simulation to real-world deployment on bimanual and single-arm platforms. Co-leading a Vision-Language-Action (VLA) research project at City Science Lab, Taipei Tech.

EDUCATION

M.S. Electrical Engineering
Taipei Tech · Sep 2025 – Present
GPA 4.0. Robotic arms, image processing, vision-/LiDAR-based navigation.

B.S. Electrical Engineering
Taipei Tech · Sep 2021 – Jun 2025
GPA 3.72. Image processing, CV data augmentation, robot navigation.

TECHNICAL SKILLS

Robot Learning
`LeRobot` `ACT` `SmolVLA` `VLA`

Robotics
`ROS / ROS2` `Franka Panda` `OpenArm` `GELLO teleop`

Simulation
`Isaac Sim` `Isaac Lab` `Isaac Lab Mimic` `Gazebo`

Navigation & Perception
`Visual SLAM` `Nav2` `RGB-D` `LiDAR` `AprilTag`

Programming
`Python` `C / C++`

LANGUAGES

Mandarin Chinese — native
English — fluent

AWARDS & ACTIVITIES

IWCE Paper Competition — Merit Award · Dec 2023

NTUT PBL International Workshop · 2023 / 2024 / 2025

RESEARCH EXPERIENCE

Research Assistant — City Science Lab, Taipei Tech
2023 – Present

- Co-leading the lab's Vision-Language-Action (VLA) project for robotic arms, from teleoperated data collection through simulation-based data generation to real-world policy deployment.
- Built an autonomous multi-map switching navigation system (ROS2 Action Server + Nav2) for indoor mobile robots.
- Designed and integrated a vision-based indoor navigation system (Visual SLAM + AprilTag localization) for a quadruped robot.

Research Assistant — InstAI (remote)
Aug 2023 – Nov 2024

- Built an end-to-end edge-AI image recognition pipeline — data capture, augmentation, model training, and deployment of custom recognition models to edge devices.

SELECTED PROJECTS

Vision-Language-Action Models on a Bimanual Robot (OpenArm)
2026

Training a VLA policy entirely in simulation and transferring it to a real bimanual arm for a pick-and-handover task. VR teleoperation (Isaac Sim) → Isaac Lab Mimic data generation → SmolVLA.

10 → ~400 demos generated 80% sim success / 100 rollouts
40% real success / 10 rollouts ~30 Hz control rate

Imitation Learning on Franka Emika Panda
2025

Leader-follower teleoperation (GELLO, Franka ROS) and ACT training pipeline; deployed on the real robot for drawer-opening and pick-and-place.

~100 demos / task ~10 Hz control rate real robot deployment

Vision-Language-Action Models on a Low-Cost Arm (Koch)
2026

SmolVLA trained on teleoperated, language-annotated demonstrations for real-time multi-task switching from natural-language instructions.

60–100 episodes / task ~30 Hz control rate

Autonomous Navigation — Quadruped
2024

Visual SLAM + AprilTag localization for indoor quadruped navigation.